ITSC 206 / LAB 5

Windows Domain & Network Services

A visual implementation report documenting segmented pfSense networks, DHCP, Windows Server 2022, Active Directory domain services, and Windows 10 client integration.

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ENVIRONMENT	VMware virtual lab
CORE SERVICES	pfSense | DHCP | AD DS | DNS

LAB OVERVIEW

What the environment demonstrates

The lab connects two pfSense appliances with separate LAN segments and validates core enterprise services on the 192.168.69.0/24 network. A Windows Server 2022 host provides the Sidhu.ca domain and DNS service, while pfSense provides DHCP and gateway services to a Windows 10 domain client.



Configuration checklist

STEP	COMPONENT	VALIDATION TARGET
01	pfSense 2 DHCP pool	192.168.69.10 - 192.168.69.100
02	Windows Server	Static IPv4 address 192.168.69.3
03	Domain services	Sidhu.ca with AD DS and DNS
04	Windows 10 client	Domain joined and DHCP configured
05	Client DNS	192.168.69.3 points to the domain server

Result

The evidence shows successful DHCP assignment, domain membership, gateway configuration, and DNS resolution settings for the Windows 10 client.

01 / EDGE SERVICES

pfSense 2: LAN and DHCP scope

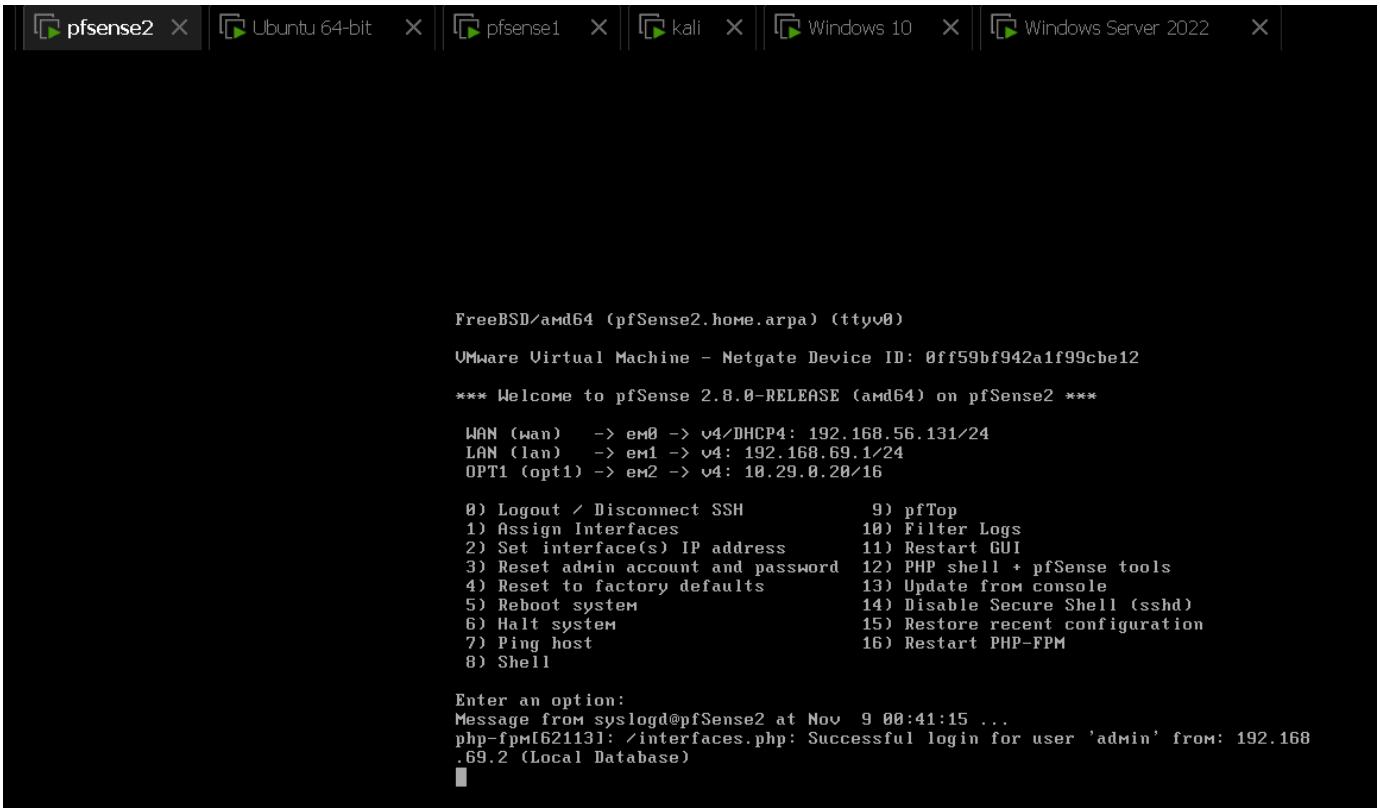


Figure 1. pfSense 2 console displaying WAN, LAN, and OPT1 interface addressing.

Interface role	pfSense 2 uses LAN address **192.168.69.1/24** and connects to the transit network through OPT1 at **10.29.0.20/16**.
Why it matters	The LAN interface acts as the default gateway for systems on the 192.168.69.0/24 subnet.

02 / ADDRESS MANAGEMENT

DHCP address pool

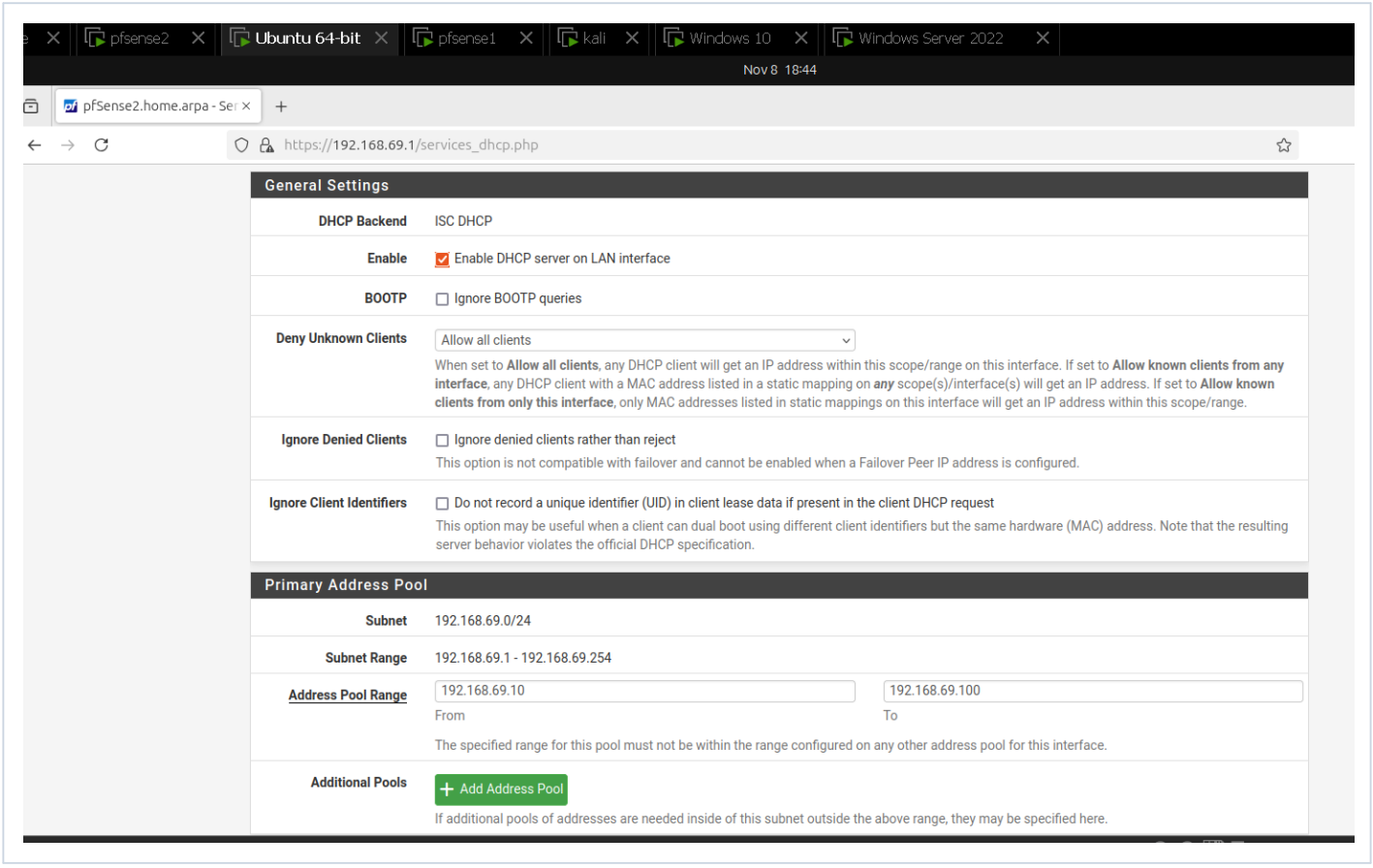


Figure 2. pfSense DHCP configuration for the 192.168.69.0/24 LAN.

Configured range	The DHCP pool is enabled from **192.168.69.10** through **192.168.69.100**.
Design choice	The server uses a lower static address outside the dynamic pool, preventing address conflicts while keeping infrastructure addresses predictable.

03 / NETWORK SEGMENTATION

pfSense 1: upstream segment

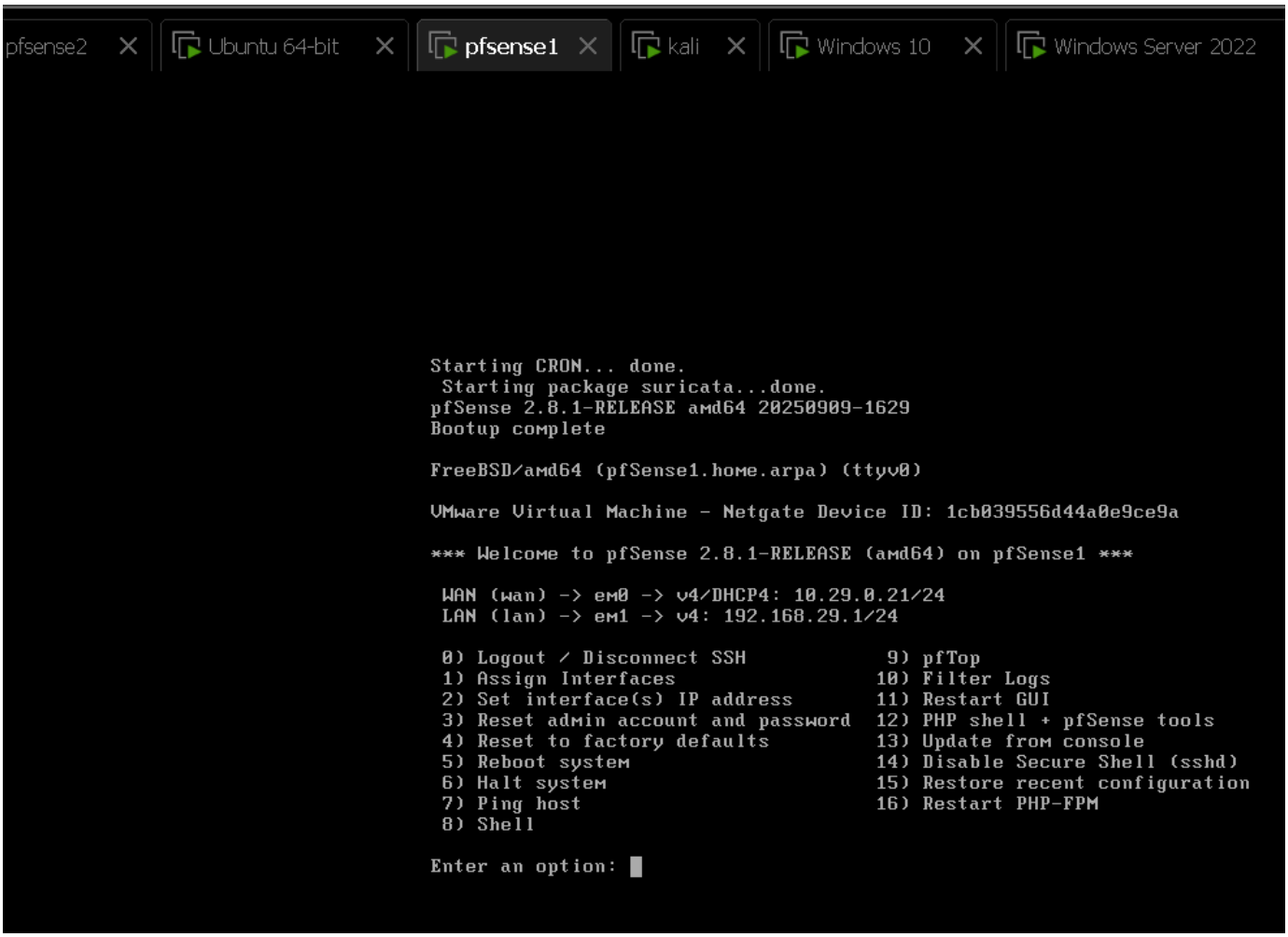


Figure 3. pfSense 1 console with WAN and LAN interface assignments.

Interface role	pfSense 1 uses **192.168.29.1/24** on its LAN and receives **10.29.0.21/24** on its WAN interface.
Validation	The second firewall documents the separate upstream segment used in the multi-router virtual topology.

04 / IDENTITY AND NAME SERVICES

Windows Server 2022 domain host

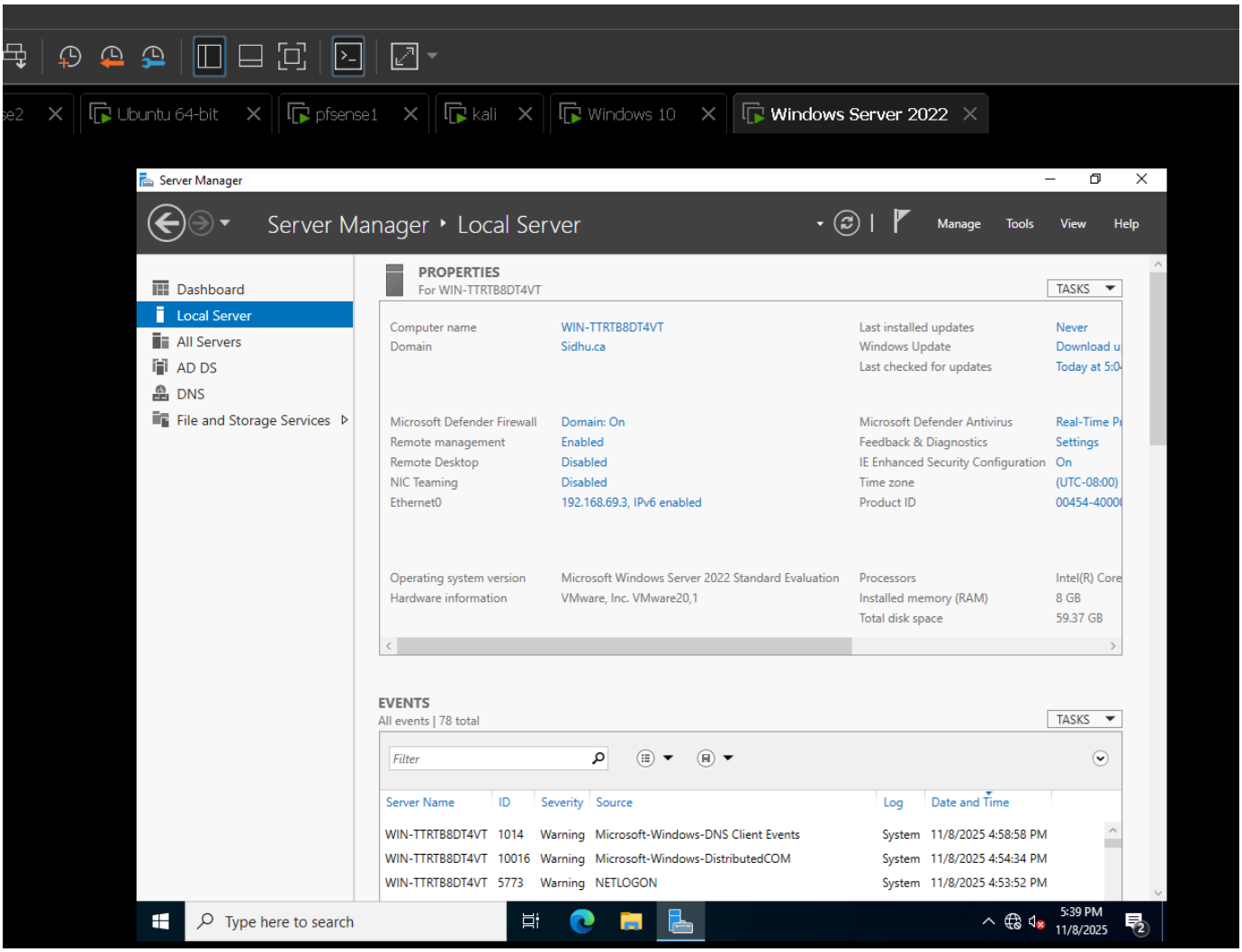


Figure 4. Windows Server 2022 Server Manager showing the Sidhu.ca domain and installed roles.

Server identity	The host is joined to the **Sidhu.ca** domain and has **AD DS** and **DNS** roles visible in Server Manager.
Network role	The server is assigned **192.168.69.3**, providing a stable endpoint for domain authentication and DNS queries.

05 / SERVER ADDRESSING

Static IPv4 configuration

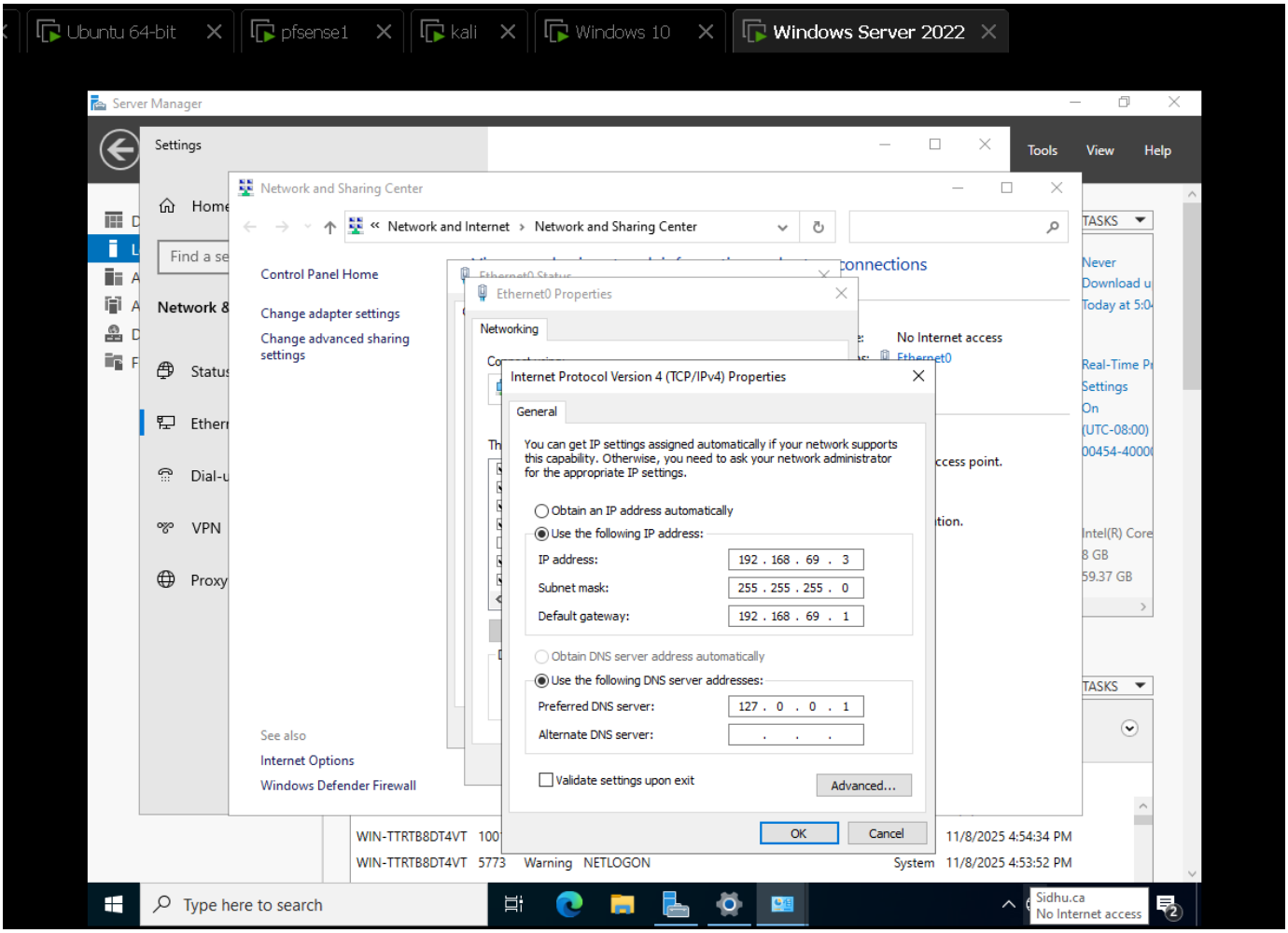


Figure 5. Windows Server IPv4 settings with a static address outside the DHCP pool.

IPv4 settings	Address **192.168.69.3**, subnet mask **255.255.255.0**, and default gateway **192.168.69.1**.
DNS setting	The server references **127.0.0.1** as its preferred DNS server, directing DNS requests to its locally installed DNS service.

06 / CLIENT INTEGRATION

Windows 10 domain client

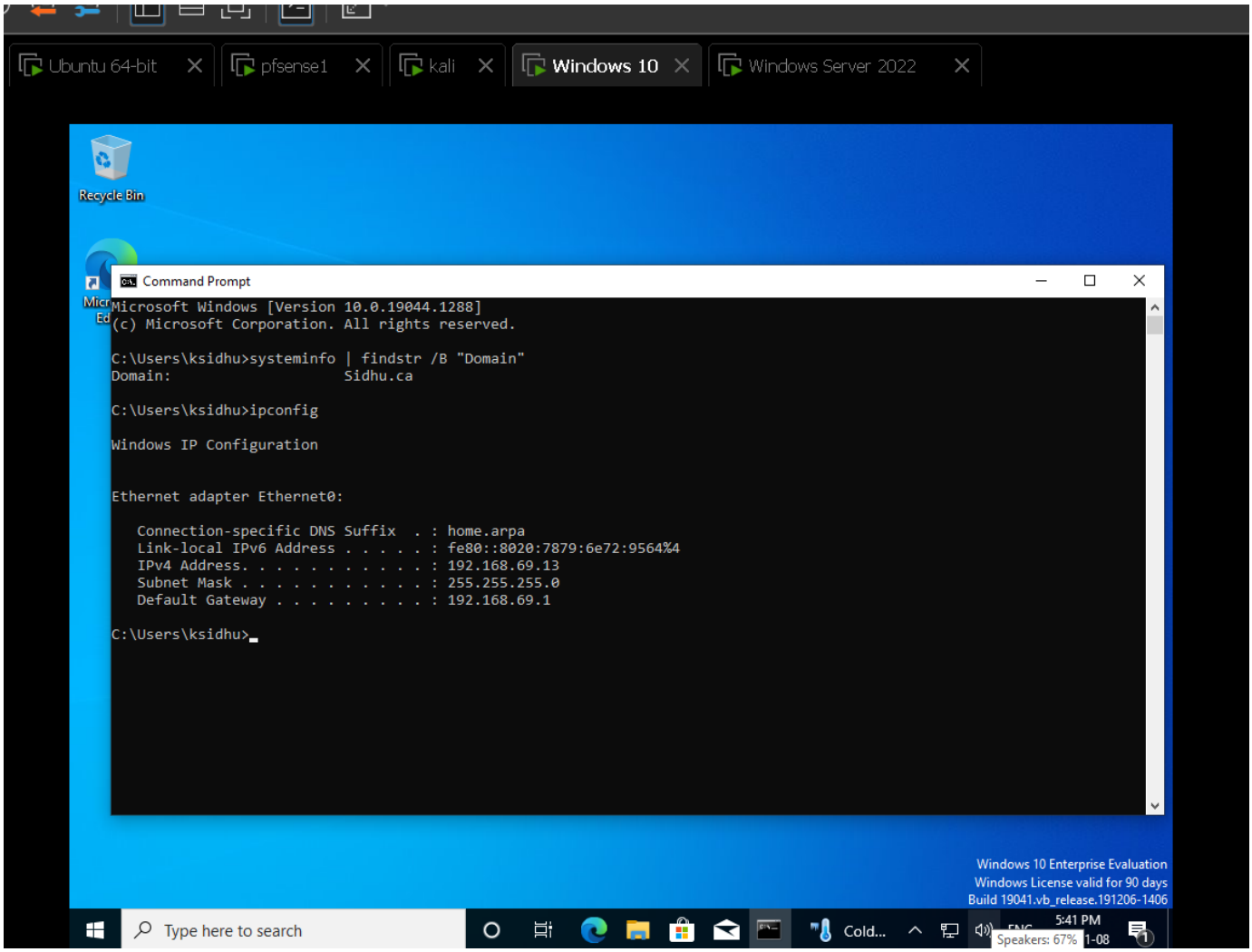


Figure 6. Windows 10 command prompt confirming domain membership and DHCP-assigned addressing.

Authentication	The system reports membership in the **Sidhu.ca** domain.
Address lease	The client receives **192.168.69.13** with gateway **192.168.69.1**, confirming successful DHCP service on the LAN.

07 / SERVICE VERIFICATION

Client DNS validation

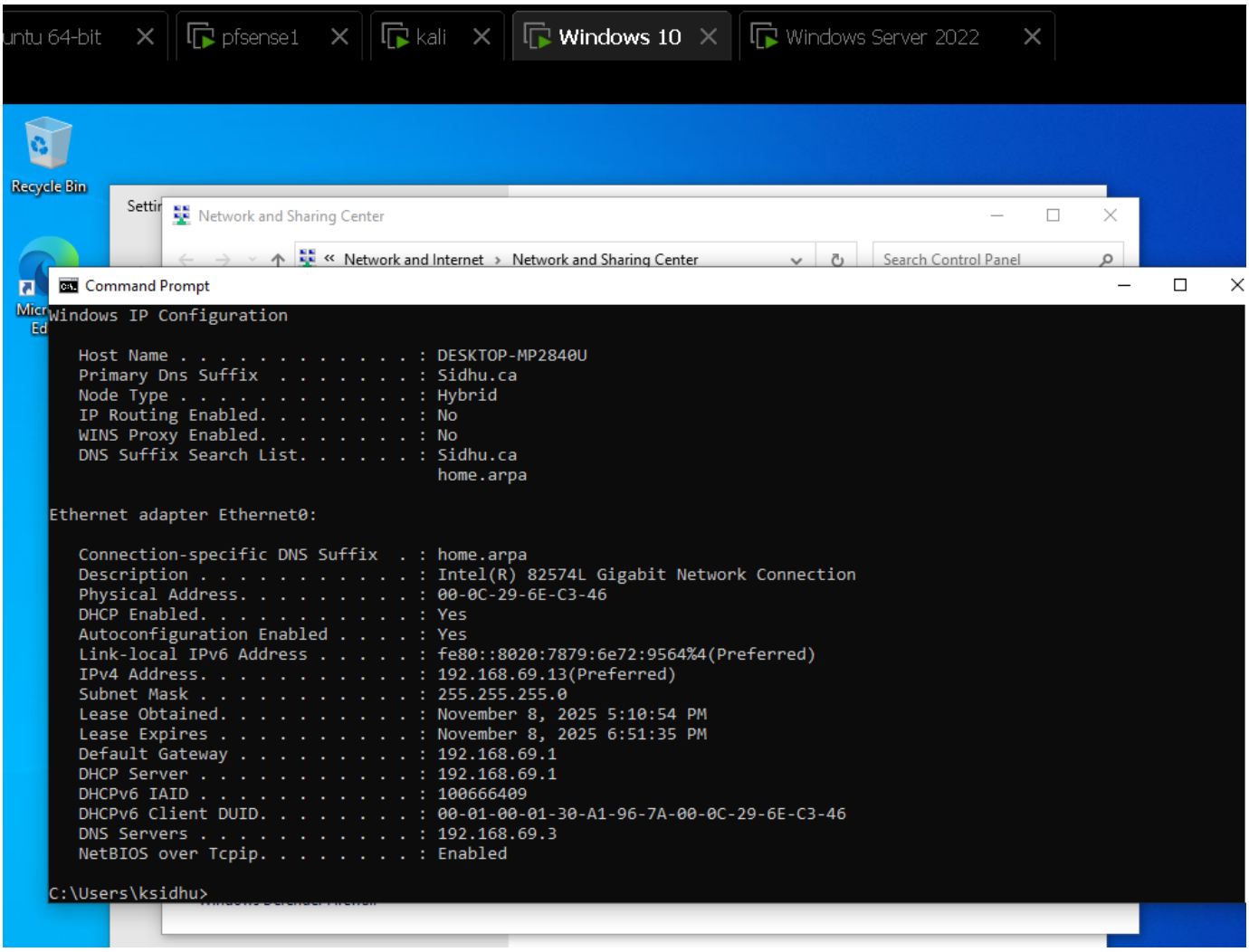


Figure 7. Windows 10 IP configuration showing the domain DNS server.

DNS path	The client uses **192.168.69.3** as its DNS server, allowing it to locate domain resources and services.
End state	DHCP, gateway assignment, DNS, and domain integration are aligned across the same 192.168.69.0/24 client network.

IMPLEMENTATION SUMMARY

Validated outcomes

Network segmentation	Two pfSense systems represent separate network segments connected through a transit network.
Address management	pfSense 2 distributes client addresses from 192.168.69.10 to 192.168.69.100.
Server infrastructure	Windows Server 2022 uses static address 192.168.69.3 and hosts AD DS and DNS for Sidhu.ca.
Client integration	Windows 10 receives 192.168.69.13 by DHCP, uses 192.168.69.1 as its gateway, and points DNS to 192.168.69.3.
Verification	Command-line output and Server Manager screenshots provide evidence of the working configuration.

Conclusion

The lab demonstrates an end-to-end virtual enterprise network configuration: segmented routing, controlled address allocation, static server addressing, centralized identity and DNS services, and successful Windows client enrollment. The evidence is organized by service so the configuration can be reviewed quickly and reproduced systematically.